Experiment 19: Planet Knowledge Representation

Aim:

To implement a knowledge base of planetary facts and retrieve information using logical inference.

Algorithm:

1. Store facts about planets (distance, moons, type).
2. Define query functions for different attributes.
3. Answer questions by retrieving matching facts.

Code:

```
planets = {
    'Mercury': {'type': 'Terrestrial', 'moons': 0, 'order': 1},
    'Venus':   {'type': 'Terrestrial', 'moons': 0, 'order': 2},
    'Earth':   {'type': 'Terrestrial', 'moons': 1, 'order': 3},
    'Mars':    {'type': 'Terrestrial', 'moons': 2, 'order': 4},
    'Jupiter': {'type': 'Gas Giant', 'moons': 95, 'order': 5},
    'Saturn':  {'type': 'Gas Giant', 'moons': 146, 'order': 6}
}

def query(planet, attr):
    return planets.get(planet, {}).get(attr, 'Unknown')

print("Earth moons:", query('Earth', 'moons'))
print("Jupiter type:", query('Jupiter', 'type'))
print("Gas giants:", [p for p,d in planets.items() if d['type']=='Gas Giant'])
```

Sample Output:

```
Earth moons: 1
Jupiter type: Gas Giant
Gas giants: ['Jupiter', 'Saturn']
```

Result:

The knowledge base successfully stored and retrieved planetary facts in response to queries.